

```
Processing triggers for libc-bin (2.35-0ubuntu3.5) ...  
nlr2007@NLR:~$ mkdir OSSP_Skill_Shell_24P  
  
cd OSSP_Skill_Shell_24P  
nlr2007@NLR:~/OSSP_Skill_Shell_24P$ |
```

Folder Creation

```
nlr2007@NLR:~/OSSP_Skill_Shell_24P$ mkdir Program_01_Setup_Process  
mkdir Program_02_Shell_Loop  
mkdir Program_03_Command_History  
mkdir Program_04_Command_Parser  
mkdir Program_05_Quote_Handler  
nlr2007@NLR:~/OSSP_Skill_Shell_24P$ |
```

Program 01:

```

#include <unistd.h>
#include <sys/wait.h>
#include <stdlib.h>

int main()
{
    pid_t pid;

    printf("OSSP Program 01\n");

    pid = fork();

    if(pid < 0)
    {
        printf("Fork failed\n");
        return 1;
    }

    else if(pid == 0)
    {
        printf("Child Process\n");
        printf("Child PID: %d\n", getpid());

        execl("/bin/ls", "ls", NULL);

        printf("Exec failed\n");
    }

    else
    {
        printf("Parent Process\n");
        printf("Parent PID: %d\n", getpid());

        waitpid(pid, NULL, 0);

        printf("Child completed\n");
    }

    return 0;
}
nlr2007@NLR:~/OSSP_Skill_Shell_24P/Program_01_Setup_Process$ |

```

Output

```

nlr2007@NLR:~/OSSP_Skill_Shell_24P/Program_01_Setup_Process$ gcc main.c -o p
1
nlr2007@NLR:~/OSSP_Skill_Shell_24P/Program_01_Setup_Process$ ./p1
OSSP Program 01
Parent Process
Parent PID: 2000
Child Process
Child PID: 2001
main.c p1
Child completed
nlr2007@NLR:~/OSSP_Skill_Shell_24P/Program_01_Setup_Process$ |

```

Program 02:

```
GNU nano 7.2                                main.c *
#include <stdio.h>
#include <string.h>

int main()
{
    char command[100];

    while(1)
    {
        printf("OSSP-Shell> ");

        fgets(command, sizeof(command), stdin);

        command[strlen(command)-1]='\0';

        if(strcmp(command,"exit")==0)
        {
            printf("Shell terminated\n");
            break;
        }

        printf("You entered: %s\n",command);
    }

    return 0;
}
```

OutPut:

```
hlr2007@NLR:~/OSSP_Skill_Shell_24P/Program_02_Shell_Loop$ gcc main.c -o p2
hlr2007@NLR:~/OSSP_Skill_Shell_24P/Program_02_Shell_Loop$ ./p2
OSSP-Shell> hello
You entered: hello
OSSP-Shell> bye
You entered: bye
OSSP-Shell> |
```

Program 03:

```
nlr2007@NLR:~/OSSP_Skill_Shell_24P/Program_03_Command_History$ cat main.c
#include <stdio.h>
#include <string.h>

int main()
{
    char history[10][100];
    char command[100];
    int count=0;

    while(1)
    {
        printf("OSSP-Shell> ");
        fgets(command,100,stdin);
        command[strlen(command)-1]='\0';

        if(strcmp(command,"exit")==0)
            break;

        if(strcmp(command,"history")==0)
        {
            for(int i=0;i<count;i++)
            {
                printf("%d %s\n",i+1,history[i]);
            }

            continue;
        }

        strcpy(history[count],command);
        count++;
    }

    return 0;
}
nlr2007@NLR:~/OSSP_Skill_Shell_24P/Program_03_Command_History$ |
```

Output:

```
1 nlr2007@NLR:~/OSSP_Skill_Shell_24P/Program_03_Command_History$ ls
main.c
nlr2007@NLR:~/OSSP_Skill_Shell_24P/Program_03_Command_History$ gcc main.c -o p3
nlr2007@NLR:~/OSSP_Skill_Shell_24P/Program_03_Command_History$ ./p3
OSSP-Shell> ls
OSSP-Shell> pwd
OSSP-Shell> exit
nlr2007@NLR:~/OSSP_Skill_Shell_24P/Program_03_Command_History$ |
```

Program 4:

```
GNU nano 7.2 main.c *
#include <stdio.h>
#include <string.h>

int main()
{
    char input[200];
    char *token;

    printf("Enter command: ");
    fgets(input,200,stdin);

    token=strtok(input," ");

    while(token!=NULL)
    {
        printf("Token: %s\n",token);
        token=strtok(NULL," ");
    }

    return 0;
}
```

Output:

```

nlr2007@NLR:~/OSSP_Skill_Shell_24P/Program_04_Command_Parser$ nano main.c
nlr2007@NLR:~/OSSP_Skill_Shell_24P/Program_04_Command_Parser$ gcc main.c -o p4
nlr2007@NLR:~/OSSP_Skill_Shell_24P/Program_04_Command_Parser$ ./p4
Enter command: ls
Token: ls

nlr2007@NLR:~/OSSP_Skill_Shell_24P/Program_04_Command_Parser$ |

```

Program 05:

```

GNU nano 7.2
#include <stdio.h>
#include <string.h>

int main()
{
    char input[200];

    printf("Enter command:");
    fgets(input,200,stdin);

    if(input[0]=='\\')
    {
        printf("Quoted string detected\n");
    }

    printf("Command: %s",input);

    return 0;
}

```

Output:

```

nlr2007@NLR:~/OSSP_Skill_Shell_24P/Program_05_Quote_Handler$ gcc main.c -o p5
nlr2007@NLR:~/OSSP_Skill_Shell_24P/Program_05_Quote_Handler$ ./p5
Enter command:ls
Command: ls
nlr2007@NLR:~/OSSP_Skill_Shell_24P/Program_05_Quote_Handler$ |

```

Program 06:

```
GNU nano 7.2 main.c
#include <stdio.h>
#include <string.h>

void remove_escape(char *input)
{
    int i,j=0;

    for(i=0; input[i]!='\0'; i++)
    {
        if(input[i]=='\\')
        {
            continue;
        }

        input[j++]=input[i];
    }

    input[j]='\0';
}

int main()
{
    char input[200];

    printf("Enter command: ");
    fgets(input,200,stdin);

    input[strlen(input)-1]='\0';

    remove_escape(input);

    printf("Processed command: %s\n",input);

    return 0;
}
```

Output:

```
nlr2007@NLR:~/OSSP_Skill_Shell_24P/Program_06_Escape_Handler$ ./p6
Enter command: ls
Processed command: ls
nlr2007@NLR:~/OSSP_Skill_Shell_24P/Program_06_Escape_Handler$ |
```

Program 06:

```
#include <stdio.h>
#include <unistd.h>
#include <sys/wait.h>
#include <stdlib.h>

int main()
{
    char command[100];

    printf("OSSP-Shell> ");
    scanf("%s", command);

    pid_t pid=fork();

    if(pid==0)
    {
        execlp(command, command, NULL);

        printf("Command not found\n");
        exit(1);
    }
    else
    {
        wait(NULL);
        printf("Command completed\n");
    }

    return 0;
}
```

Output:

```
nlr2007@NLR:~/OSSP_Skill_Shell_24P/Program_07_Process_Execution$ gcc main.c -o p7
nlr2007@NLR:~/OSSP_Skill_Shell_24P/Program_07_Process_Execution$ ./p7
OSSP-Shell> ls
main.c  p7
Command completed
nlr2007@NLR:~/OSSP_Skill_Shell_24P/Program_07_Process_Execution$ |
```

Program 08:

```
#include <stdio.h>
#include <stdlib.h>
#include <string.h>

int main()
{
    char *path=getenv("PATH");

    printf("System PATH:\n");
    printf("%s\n",path);

    char *token=strtok(path,":");

    while(token!=NULL)
    {
        printf("Directory: %s\n",token);

        token=strtok(NULL,":");
    }

    return 0;
}
```

Output:

```
Directory: /sbin
Directory: /bin
Directory: /usr/games
Directory: /usr/local/games
Directory: /usr/lib/wsl/lib
Directory: /mnt/c/Program Files/Microsoft/jdk-21.0.12.101-hotspot/bin
Directory: /mnt/c/Program Files/Common Files/Oracle/Java/javapath
Directory: /mnt/c/Program Files (x86)/Common Files/Oracle/Java/java8path
Directory: /mnt/c/Program Files (x86)/Common Files/Oracle/Java/javapath
Directory: /mnt/c/WINDOWS/system32
Directory: /mnt/c/WINDOWS
Directory: /mnt/c/WINDOWS/System32/Wbem
Directory: /mnt/c/WINDOWS/System32/WindowsPowerShell/v1.0/
Directory: /mnt/c/WINDOWS/System32/OpenSSH/
Directory: /mnt/c/Program Files/dotnet/
Directory: /mnt/c/MinGW/bin
Directory: /mnt/c/Users/HP/Downloads/Code Downloads
Directory: /mnt/c/Program Files/Java/jdk-25/bin
Directory: /mnt/c/Program Files/Git/bin
```

Program 09:

```

#include <stdio.h>
#include <stdlib.h>
#include <string.h>

int main()
{
    char variable[50];

    printf("Enter variable name: ");
    scanf("%s",variable);

    char *value=getenv(variable);

    if(value)
    {
        printf("Value: %s\n",value);
    }

    else
    {
        printf("Variable not found\n");
    }

    return 0;
}

```

Output:

```

nrlr2007@NLR:~/OSSP_Skill_Shell_24P/Program_09_Variable_Expansion$ nano main.c
nrlr2007@NLR:~/OSSP_Skill_Shell_24P/Program_09_Variable_Expansion$ gcc main.c -o p9
nrlr2007@NLR:~/OSSP_Skill_Shell_24P/Program_09_Variable_Expansion$ ./p9
Enter variable name: a
Variable not found
nrlr2007@NLR:~/OSSP_Skill_Shell_24P/Program_09_Variable_Expansion$ ./p9
Enter variable name: Home
Variable not found
nrlr2007@NLR:~/OSSP_Skill_Shell_24P/Program_09_Variable_Expansion$ |

```

Program 10:

```
GNU nano 7.2
#include <stdio.h>
#include <unistd.h>
#include <string.h>
#include <stdlib.h>

int main()
{
    char command[100];

    while(1)
    {
        printf("OSSP-Shell> ");
        scanf("%s",command);

        if(strcmp(command,"exit")==0)
        {
            break;
        }

        else if(strcmp(command,"pwd")==0)
        {
            char cwd[200];
            getcwd(cwd,sizeof(cwd));
            printf("%s\n",cwd);
        }

        else if(strcmp(command,"cd")==0)
        {
            char path[100];
            scanf("%s",path);

            if(chdir(path)!=0)
            {
                printf("Directory error\n");
            }
        }

        else
        {
            printf("Unknown command\n");
        }
    }

    return 0;
}
```

Output:

```
nlr2007@NLR:~/OSSP_Skill_Shell_24P/Program_10_Builtin_Commands$ gcc main.c -o p10
nlr2007@NLR:~/OSSP_Skill_Shell_24P/Program_10_Builtin_Commands$ ./p10
OSSP-Shell> pwd
/home/nlr2007/OSSP_Skill_Shell_24P/Program_10_Builtin_Commands
OSSP-Shell> cd /tmp
OSSP-Shell> pwd
/tmp
OSSP-Shell> |
```

Program 11:

```

GNU nano 7.2
#include <stdio.h>
#include <string.h>

#define SIZE 10

char history[SIZE][100];

int count = 0;

void add_history(char *cmd)
{
    if(count < SIZE)
    {
        strcpy(history[count],cmd);
        count++;
    }
}

void show_history()
{
    int i;

    for(i=0;i<count;i++)
    {
        printf("%d : %s\n",i+1,history[i]);
    }
}

int main()
{
    char command[100];

    while(1)
    {
        printf("OSSP-Shell> ");
        fgets(command,100,stdin);
        command[strlen(command)-1]='\0';

        if(strcmp(command,"exit")==0)
            break;

        if(strcmp(command,"history")==0)
        {
            show_history();
            continue;
        }

        add_history(command);
        printf("Executed : %s\n",command);
    }

    return 0;
}

```

Output:

```
nlr2007@NLR:~/OSSP_Skill_Shell_24P/Program_11_History_Buffer$ gcc main.c -o p11
nlr2007@NLR:~/OSSP_Skill_Shell_24P/Program_11_History_Buffer$ ./p11
OSSP-Shell>
```

Program 11:

```
#include <stdio.h>
#include <unistd.h>
#include <sys/wait.h>

int main()
{
    int pipefd[2];
    pipe(pipefd);

    pid_t pid=fork();

    if(pid==0)
    {
        dup2(pipefd[1],STDOUT_FILENO);
        close(pipefd[0]);

        execlp("ls","ls",NULL);
    }

    else
    {
        wait(NULL);

        dup2(pipefd[0],STDIN_FILENO);
        close(pipefd[1]);

        execlp("grep","grep",".c",NULL);
    }

    return 0;
}
```

Output:

```

nlr2007@NLR:~/OSSP_Skill_Shell_24P/Program_12_Pipe$ gcc main.c -o p12
nlr2007@NLR:~/OSSP_Skill_Shell_24P/Program_12_Pipe$ ./p12
main.c
nlr2007@NLR:~/OSSP_Skill_Shell_24P/Program_12_Pipe$ |

```

Program 13:

```

#include <stdio.h>
#include <fcntl.h>
#include <unistd.h>

int main()
{
    int fd;

    fd=open("input.txt",O_RDONLY);

    dup2(fd,STDIN_FILENO);

    close(fd);

    execlp("cat","cat",NULL);

    return 0;
}

```

Output:

```

nlr2007@NLR:~/OSSP_Skill_Shell_24P/Program_12_Pipe$ mkdir Program_13_Input_Redirection
nlr2007@NLR:~/OSSP_Skill_Shell_24P/Program_12_Pipe$ cd Program_13_Input_Redirection
nlr2007@NLR:~/OSSP_Skill_Shell_24P/Program_12_Pipe/Program_13_Input_Redirection$ nano main.c
" > input.txt
nlr2007@NLR:~/OSSP_Skill_Shell_24P/Program_12_Pipe/Program_13_Input_Redirection$ gcc main.c -o input_r
edirect
nlr2007@NLR:~/OSSP_Skill_Shell_24P/Program_12_Pipe/Program_13_Input_Redirection$ ./input_redirect
OSSP Input File
nlr2007@NLR:~/OSSP_Skill_Shell_24P/Program_12_Pipe/Program_13_Input_Redirection$ |

```

Program 14:

```
#include <stdio.h>
#include <fcntl.h>
#include <unistd.h>

int main()
{

int fd;

fd=open("output.txt",
O_WRONLY|O_CREAT|O_TRUNC,
0644);

dup2(fd,STDOUT_FILENO);

close(fd);

printf("Output redirected successfully\n");

return 0;

}|
```

Output:

```
nlr2007@NLR:~/OSSP_Skill_Shell_24P/Program_14_Output_Redirection$ ./output_redirect
nlr2007@NLR:~/OSSP_Skill_Shell_24P/Program_14_Output_Redirection$ cat output.txt
Output redirected successfully
nlr2007@NLR:~/OSSP_Skill_Shell_24P/Program_14_Output_Redirection$ |
```

Program 15:

```
#include <stdio.h>
#include <fcntl.h>
#include <unistd.h>

int main()
{

int fd;

fd=open("log.txt",
O_WRONLY|O_CREAT|O_APPEND,
0644);

dup2(fd,STDOUT_FILENO);

close(fd);

printf("New log entry\n");

int errorfd;

errorfd=open("error.txt",
O_WRONLY|O_CREAT|O_TRUNC,
0644);

dup2(errorfd,STDERR_FILENO);

close(errorfd);

fprintf(stderr,
"Sample error message\n");

return 0;
```

Output:

```
nlr2007@NLR:~/OSSP_Skill_Shell_24P/Program_15_Advanced_Redirection$ nano main.c
nlr2007@NLR:~/OSSP_Skill_Shell_24P/Program_15_Advanced_Redirection$ gcc main.c -o redirect
nlr2007@NLR:~/OSSP_Skill_Shell_24P/Program_15_Advanced_Redirection$ ./redirect
nlr2007@NLR:~/OSSP_Skill_Shell_24P/Program_15_Advanced_Redirection$ cat log.tx
cat: log.tx: No such file or directory
 1 nlr2007@NLR:~/OSSP_Skill_Shell_24P/Program_15_Advanced_Redirection$ cat log.txt
New log entry
nlr2007@NLR:~/OSSP_Skill_Shell_24P/Program_15_Advanced_Redirection$ cat error.txt
Sample error message
nlr2007@NLR:~/OSSP_Skill_Shell_24P/Program_15_Advanced_Redirection$ |
```

Program 16:

```
GNU nano 7.2 main.c *
#include <stdio.h>
#include <unistd.h>
#include <stdlib.h>
#include <sys/wait.h>

int main()
{
    char command[50];

    printf("OSSP-Shell> ");
    scanf("%s",command);

    pid_t pid=fork();

    if(pid==0)
    {
        printf("Running background process PID: %d\n",
getpid());

        execlp(command,command,NULL);
        exit(1);
    }

    else
    {
        printf("Shell returned immediately\n");
    }

    return 0;
}
```

Output:

```
nlr2007@NLR:~/OSSP_Skill_Shell_24P/Program_16_Background_Jobs$ nano main.c
nlr2007@NLR:~/OSSP_Skill_Shell_24P/Program_16_Background_Jobs$ gcc main.c -o background
nlr2007@NLR:~/OSSP_Skill_Shell_24P/Program_16_Background_Jobs$ ./background
OSSP-Shell> sleep
Shell returned immediately
Running background process PID: 2777
nlr2007@NLR:~/OSSP_Skill_Shell_24P/Program_16_Background_Jobs$ sleep: missing operand
Try 'sleep --help' for more information.
```

Program 17:

```
#include <stdio.h>

struct Job
{
    int id;
    int pid;
    char status[20];
};

int main()
{
    struct Job jobs[5];

    jobs[0].id=1;
    jobs[0].pid=1234;
    sprintf(jobs[0].status, "Running");

    printf("Job List\n");
    printf("ID PID STATUS\n");

    for(int i=0; i<1; i++)
    {
        printf("%d %d %s\n",
            jobs[i].id,
            jobs[i].pid,
            jobs[i].status);
    }

    return 0;
}
```

Output:

```
nlr2007@NLR:~/OSSP_Skill_Shell_24P$ cd Program_17_Job_Table
nlr2007@NLR:~/OSSP_Skill_Shell_24P/Program_17_Job_Table$ nano main.c
nlr2007@NLR:~/OSSP_Skill_Shell_24P/Program_17_Job_Table$ gcc main.c -o jobs
nlr2007@NLR:~/OSSP_Skill_Shell_24P/Program_17_Job_Table$ ./jobs
Job List
ID PID STATUS
1 1234 Running
nlr2007@NLR:~/OSSP_Skill_Shell_24P/Program_17_Job_Table$ |
```

Program 18:

```
GNU nano 7.2
#include <stdio.h>
#include <signal.h>
#include <unistd.h>

void handler(int sig)
{
    printf("\nSignal received: %d\n",sig);
}

int main()
{
    signal(SIGINT,handler);

    while(1)
    {
        printf("Running...\n");
        sleep(2);
    }

    return 0;
}
```

Output:

```
nlr2007@NLR:~/OSSP_Skill_Shell_24P$ mkdir Program_18_Signal_Basics
nlr2007@NLR:~/OSSP_Skill_Shell_24P$ cd Program_18_Signal_Basics
nlr2007@NLR:~/OSSP_Skill_Shell_24P/Program_18_Signal_Basics$ nano main.c
nlr2007@NLR:~/OSSP_Skill_Shell_24P/Program_18_Signal_Basics$ gcc main.c -o signal_test
nlr2007@NLR:~/OSSP_Skill_Shell_24P/Program_18_Signal_Basics$ ./signal_test
Running...
Running...
Running...
^C
Signal received: 2
```

Program 19:

```
#include <stdio.h>
#include <signal.h>
#include <unistd.h>

void sigint_handler(int sig)
{
    printf("\nCannot terminate shell\n");
}

int main()
{
    signal(SIGINT, sigint_handler);

    while(1)
    {
        printf("OSSP-Shell Running\n");
        sleep(2);
    }

    return 0;
}
```

Output:

```

[1]+  Stopped                  ./signal_test
148 nlr2007@NLR:~/OSSP_Skill_Shell_24P/Program_18_Signal_Basics$ cd ..
nlr2007@NLR:~/OSSP_Skill_Shell_24P$ mkdir Program_19_SIGINT_Handler
nlr2007@NLR:~/OSSP_Skill_Shell_24P$ cd Program_19_SIGINT_Handler
nlr2007@NLR:~/OSSP_Skill_Shell_24P/Program_19_SIGINT_Handler$ nano main.c
nlr2007@NLR:~/OSSP_Skill_Shell_24P/Program_19_SIGINT_Handler$ gcc main.c -o sigint
nlr2007@NLR:~/OSSP_Skill_Shell_24P/Program_19_SIGINT_Handler$ ./sigint
OSSP-Shell Running
OSSP-Shell Running
^C
Cannot terminate shell
OSSP-Shell Running

```

Program 20:

```

GNU nano 7.2
#include <stdio.h>
#include <signal.h>
#include <unistd.h>

void stop_handler(int sig)
{
    printf("\nProcess stopped signal: %d\n",sig);
}

int main()
{
    signal(SIGTSTP,stop_handler);

    while(1)
    {
        printf("Process active\n");
        sleep(2);
    }

    return 0;
}

```

Output:

```
214 ^C stopped ./sigint
148 nlr2007@NLR:~/OSSP_Skill_Shell_24P/Program_19_SIGINT_Handler$ cd ..
nlr2007@NLR:~/OSSP_Skill_Shell_24P$ mkdir Program_20_SIGTSTP_Handler
nlr2007@NLR:~/OSSP_Skill_Shell_24P$ cd Program_20_SIGTSTP_Handler
nlr2007@NLR:~/OSSP_Skill_Shell_24P/Program_20_SIGTSTP_Handler$ nano main.c
nlr2007@NLR:~/OSSP_Skill_Shell_24P/Program_20_SIGTSTP_Handler$ gcc main.c -o sigtstp
nlr2007@NLR:~/OSSP_Skill_Shell_24P/Program_20_SIGTSTP_Handler$ ./sigtstp
process active
process active
Z
process stopped signal: 20
process active
```

Program 21:

```
#include <stdio.h>

void parser()
{
    printf("Parser module loaded\n");
}

void executor()
{
    printf("Executor module loaded\n");
}

void history()
{
    printf("History module loaded\n");
}

int main()
{
    printf("OSSP Shell Integration Test\n");

    parser();
    history();
    executor();

    printf("All modules connected successfully\n");

    return 0;
}
```

Output:

```

nlr2007@NLR:~/OSSP_Skill_Shell_24P$ mkdir Program_21_Module_Integration
nlr2007@NLR:~/OSSP_Skill_Shell_24P$ cd Program_21_Module_Integration
nlr2007@NLR:~/OSSP_Skill_Shell_24P/Program_21_Module_Integration$ nano main.c
nlr2007@NLR:~/OSSP_Skill_Shell_24P/Program_21_Module_Integration$ gcc main.c -o integration
nlr2007@NLR:~/OSSP_Skill_Shell_24P/Program_21_Module_Integration$ ./integration
OSSP Shell Integration Test
Parser module loaded
History module loaded
Executor module loaded
All modules connected successfully

```

Program 22:

```

#include <stdio.h>
#include <stdlib.h>

int main()
{
    char *memory;

    memory = malloc(100);

    if(memory)
    {
        printf("Memory allocated\n");
    }

    free(memory);

    printf("Memory released\n");

    return 0;
}

```

Output:

```

nlr2007@NLR:~/OSSP_Skill_Shell_24P$ cd Program_22_Valgrind_Test
nlr2007@NLR:~/OSSP_Skill_Shell_24P/Program_22_Valgrind_Test$ nano main.c
nlr2007@NLR:~/OSSP_Skill_Shell_24P/Program_22_Valgrind_Test$ gcc main.c -o memory_test~
nlr2007@NLR:~/OSSP_Skill_Shell_24P/Program_22_Valgrind_Test$ valgrind ./memory_test
valgrind: ./memory_test: No such file or directory
127 nlr2007@NLR:~/OSSP_Skill_Shell_24P/Program_22_Valgrind_Test$ nano main.c
nlr2007@NLR:~/OSSP_Skill_Shell_24P/Program_22_Valgrind_Test$ valgrind ./memory_test
valgrind: ./memory_test: No such file or directory
127 nlr2007@NLR:~/OSSP_Skill_Shell_24P/Program_22_Valgrind_Test$ |

```

Program 23:

```
GNU nano 7.2                                main.c *
#include <stdio.h>
#include <time.h>

int main()
{
    clock_t start,end;

    start=clock();

    for(int i=0;i<1000000;i++)
    {

    }

    end=clock();

    double time_taken;

    time_taken=(double)(end-start)/CLOCKS_PER_SEC;

    printf("Execution time: %lf seconds\n",
    time_taken);

    return 0;
}
```

Output:

```
nlr2007@NLR:~/OSSP_Skill_Shell_24P/Program_23_Performance_Test$ nano main.c
nlr2007@NLR:~/OSSP_Skill_Shell_24P/Program_23_Performance_Test$ gcc main.c -o performance
nlr2007@NLR:~/OSSP_Skill_Shell_24P/Program_23_Performance_Test$ ./performance
Execution time: 0.000760 seconds
nlr2007@NLR:~/OSSP_Skill_Shell_24P/Program_23_Performance_Test$ |
```

Program 24:

```

GNU nano 7.2
#include <stdio.h>
#include <string.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/wait.h>

int main()
{
    char command[100];

    while(1)
    {
        printf("OSSP-Shell> ");

        fgets(command,100,stdin);

        command[strlen(command)-1]='\0';

        if(strcmp(command,"exit")==0)
        {
            printf("Shell closed\n");
            break;
        }

        pid_t pid=fork();

        if(pid==0)
        {
            execlp(command,command,MKL);

            printf("Command not found\n");

            exit(1);
        }
        else
        {
            wait(MKL);
        }
    }

    return 0;
}

```

Output:

```

nlr2007@MLB:~/OSSP_Skill_Shell_240$ cd Program_25_Complete_Shell
nlr2007@MLB:~/OSSP_Skill_Shell_240/Program_25_Complete_Shell$ nano main.c
nlr2007@MLB:~/OSSP_Skill_Shell_240/Program_25_Complete_Shell$ gcc main.c -o ossp_shell
nlr2007@MLB:~/OSSP_Skill_Shell_240/Program_25_Complete_Shell$ ./ossp_shell
OSSP-Shell> psd
/home/nlr2007/OSSP_Skill_Shell_240/Program_25_Complete_Shell
OSSP-Shell> ls
main.c ossp_shell
OSSP-Shell> |

```